EAST TEXAS A&M
UNIVERSITY

CSCI 352.61E
DIGITAL FORENSICS

System Forensics, Investigation, and Response

Lesson 9

Linux Forensics

Learning Objective

- Summarize various types of digital forensics.

Key Concepts

- Linux file systems
- What to look for in Linux system logs
- Forensically interesting Linux directories
- Important Linux shell commands
- How to undelete files from Linux

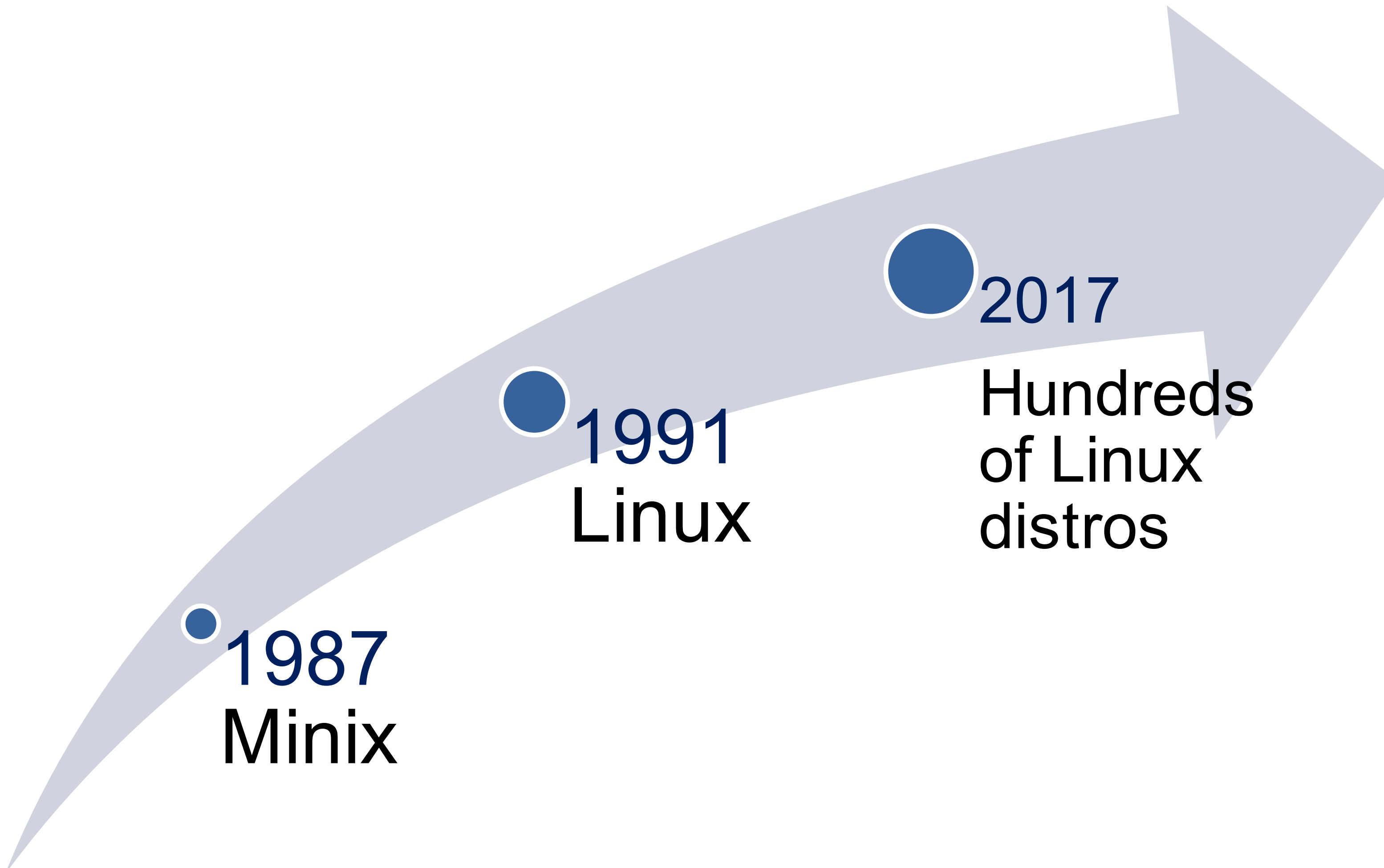
History of Linux

1969
UNIX created

1972
UNIX released

1983
GNU

History of Linux (Cont.)



Linux Shells

- Bourne shell (sh)
- Bourne-again shell (Bash)
- C shell (csh)
- Korn shell (ksh)

Common Linux Shell Commands

LINUX COMMAND	EXPLANATION AND EXAMPLE
`ls`	The `ls` command lists the contents of the current directory. Example: `ls`
`cp`	The `cp` command copies one file to another directory. Example: `cp filename.txt directoryname`
`mkdir`	The `mkdir` command creates a new directory. Example: `mkdir directoryname`
`cd`	The `cd` command is used to change directories. Example: `cd directoryname`
`rm`	The `rm` command is used to delete or remove a file. Example: `rm filename`

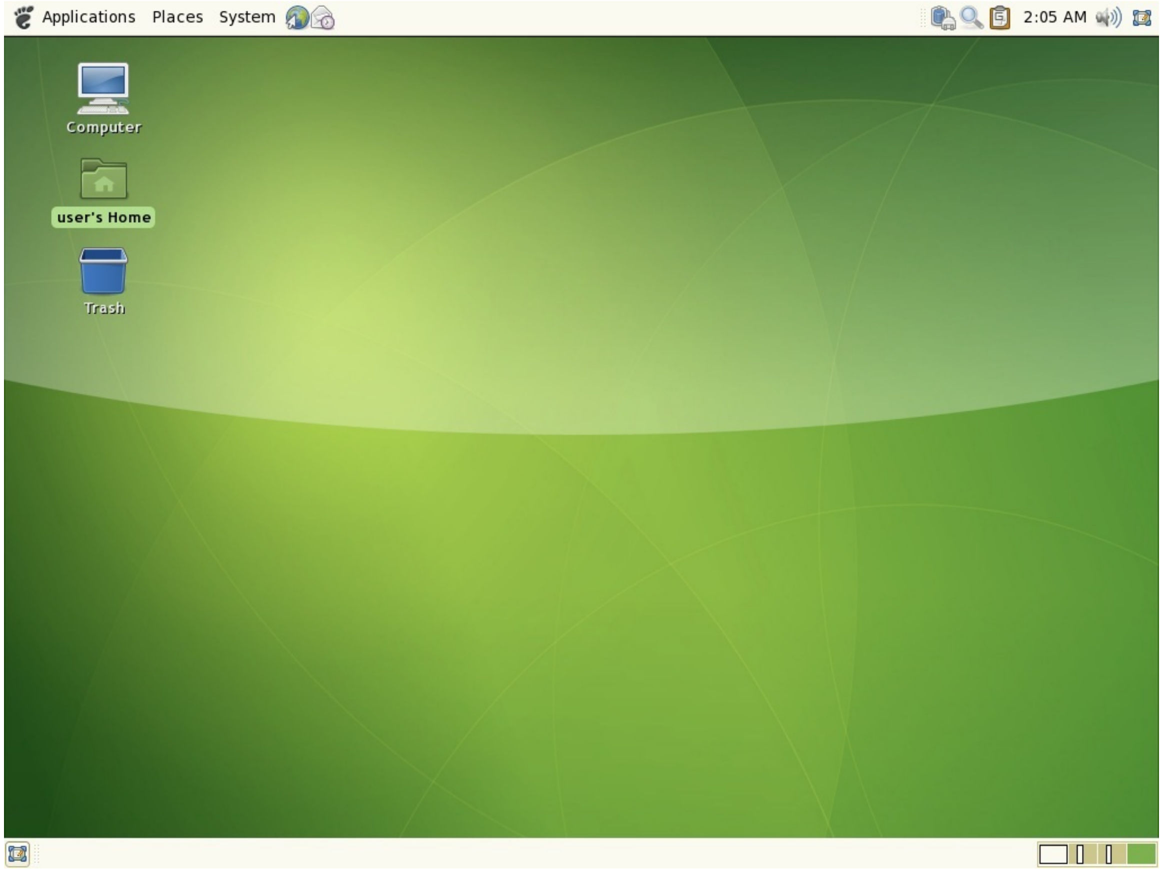
Common Linux Shell Commands (Cont.)

`rmdir`	The `rmdir` command is used to remove or delete entire directories. Example: `rmdir directoryname`
`mv`	The `mv` command is used to move a file. Example: `mv myfile.txt myfolder`
`diff`	The `diff` command performs a byte-by-byte comparison of two files and tells you what is different about them. Example: `diff myfile.txt myfile2.txt`
`cmp`	The `cmp` command performs a textual comparison of two files and tells you the difference between the two. Example: `cmp myfile.txt myfile2.txt`
>	This is the redirect command. Instead of displaying the output of a command like `ls` to the screen, it redirects it to a file. Example: `ls > file1.txt`

Common Linux Shell Commands (Cont.)

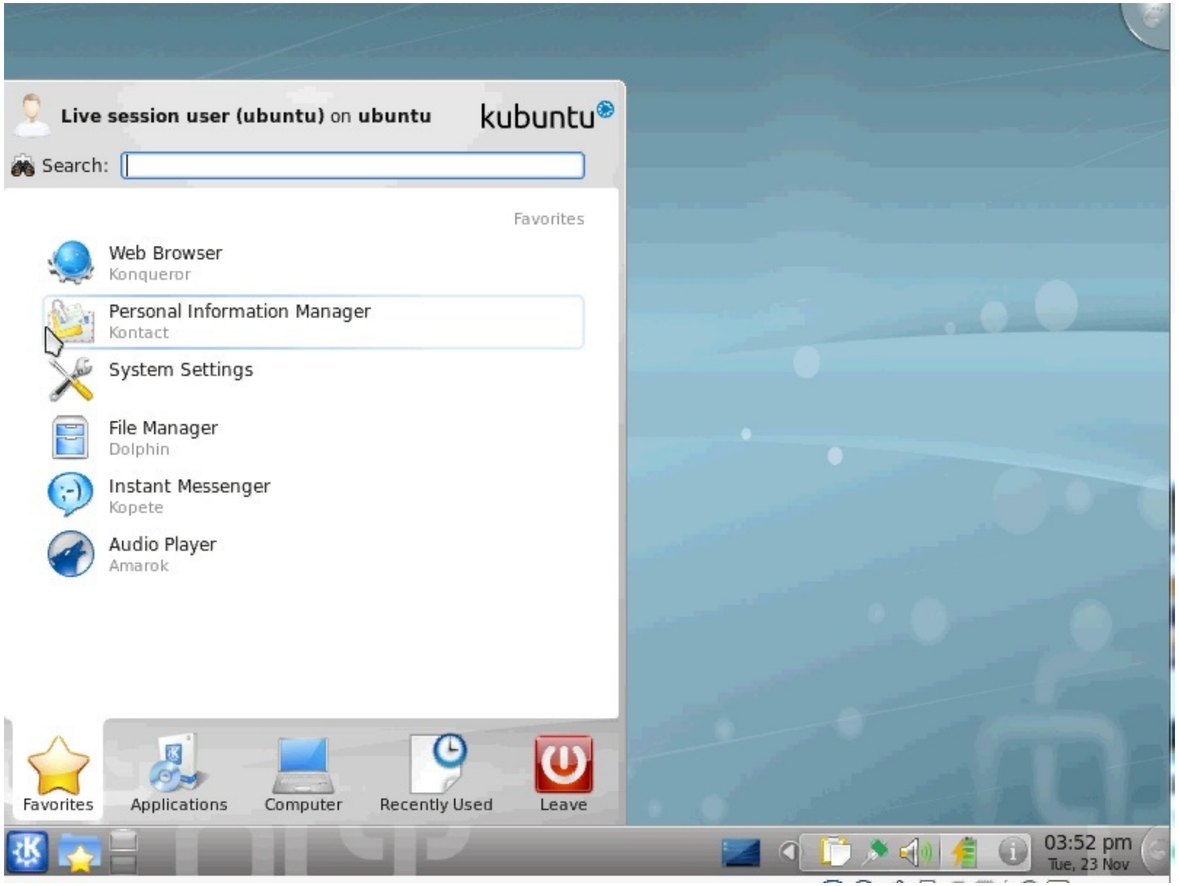
`ps`	The `ps` command lists all currently running processes that the user has started. Any program or daemon is a process. Example: `ps`
`top`	The `top` command lists all currently running processes, whether the user started them or not. It also lists more detail on the processes. Example: `top`
`fsck`	This is a file system check. The `fsck` command can check to see whether a given partition is in good working condition. Example: `fsck /dev/hda1`
`fdisk`	The `fdisk` command lists the various partitions. Example: `fdisk-1`
mount	The `mount` command mounts a partition, allowing you to work with it. Example: `mount /dev/fd0/mnt/floppy`

GNU Network Object Model Environment (GNOME)



Courtesy of The GNOME Project

K Desktop Environment (KDE)/Plasma



Courtesy of TKDE

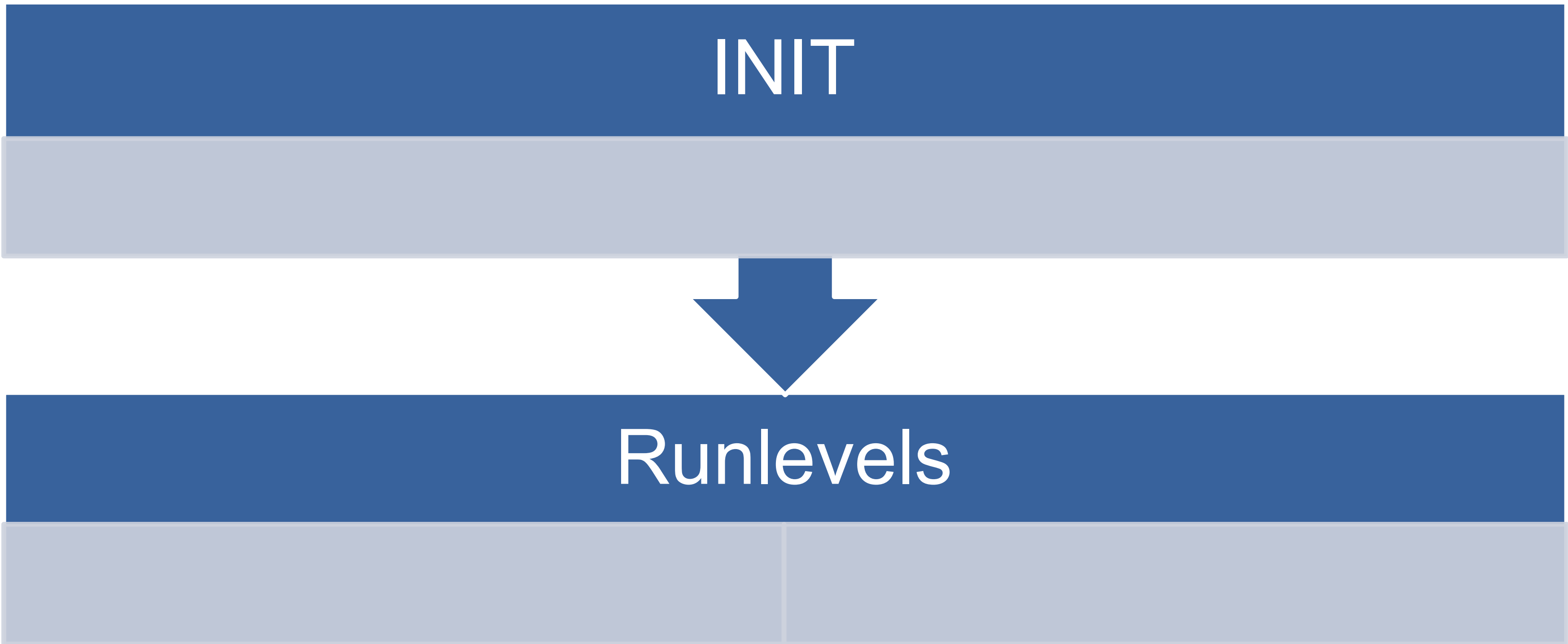
Other Linux GUIs

- Common Desktop Environment (CDE)
  - Originally developed in 1994 for UNIX systems
  - Based on HP’s Visual User Environment (VUE)
- Enlightenment
  - Relatively new
  - Designed for graphics developers

Linux Boot Process

- **BIOS**
  - POST
- **MBR**
  - GRUB
  - LILO
- **Kernel**
  - Initializes devices
  - Real mode to protected mode

Linux Boot Process (Cont.)



Run Levels

MODE	DIRECTORY	RUN LEVEL DESCRIPTION
0	/etc/rc.d/rc0.d	Halt
1	/etc/rc.d/rc1.d	Single-user mode
2	/etc/rc.d/rc2.d	Not used (user-definable)
3	/etc/rc.d/rc3.d	Full multiuser mode without GUI
4	/etc/rc.d/rc4.d	Not used (user-definable)
5	/etc/rc.d/rc5.d	Full multiuser mode with GUI
6	/etc/rc.d/rc6.d	Reboot

Logical Volume Manager

- An abstraction layer that provides volume management for the Linux kernel
- On a single system (like a single desktop or server), primary role is to allow:
  - The resizing of partitions
  - The creation of backups by taking snapshots of the logical volumes

Linux Distributions

- Open source operating system
- Popular distributions:
  - Ubuntu
  - Red Hat Enterprise Linux (RHEL)
  - OpenSUSE
  - Debian
  - Slackware

Linux File Systems

- Extended File System (ext)
  - Current version is 4
- ext4 supports volumes up to 1 exabyte and single files up to 16 terabytes
- ext3 and ext4 support three types of journaling:
  - journal (most secure)
  - ordered
  - writeback (least secure)

Linux File Systems (Cont.)

- Reiser File System
  - Supports journaling
  - Performs well when hard disk has large number of smaller files
- Berkeley Fast File System
  - Also known as UNIX File System
  - Developed at UC-Berkeley for Linux
  - Uses a bitmap to track free clusters, indicating availability

Linux Logs

Log	Contents
/var/log/faillog	Failed user logins
/var/log/kern.log	Messages from the operating system's kernel
/var/log/lpr.log	Items that have been printed
/var/log/mail.*	Email activity
/var/log/mysql.*	MySQL database server activity

Linux Logs (Cont.)

Log	Contents
/var/log/apache2/*	Apache web server activity
/var/log/lighttpd/*	Lighttpd web server activity
/var/log/apport.log	Application crashes
Intrusion detection system logs	Suspicious traffic

Viewing Logs

- Text editor in GUI
- Any of these commands work from the shell:
  - `dmesg | lpr`
  - `# tail -f /var/log/lpr.log`
  - `# less /var/log/ lpr.log`
  - `# more -f /var/log/ lpr.log`

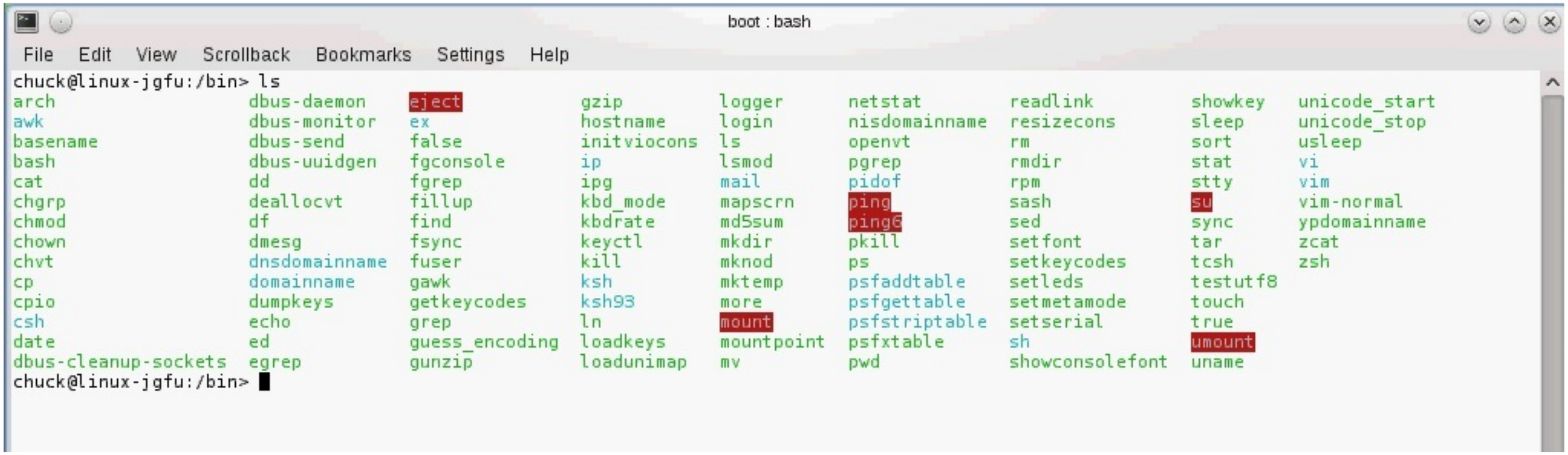
Linux Directories

- Key directories are important to the functioning of every operating system
- Directories are also important places to seek out evidence in an investigation

/root

- Home directory for the root user
  - Contains data for the administrator
- Linux root user is equivalent to Windows Administrator

The /bin Directory



boot : bash

File Edit View Scrollback Bookmarks Settings Help

```
chuck@linux-jgfu:/bin> ls
arch                  dbus-daemon     eject           gzip          logger      netstat         readlink          showkey     unicode_start
awk                   dbus-monitor    ex              hostname      login       nisdomainname   resizecons        sleep       unicode_stop
basename              dbus-send       false           initviocons   ls          openvt          rm                sort        usleep
bash                  dbus-uuidgen    fgconsole       ip            lsmod       pgrep           rmdir             stat        vi
cat                   dd              fgrep           ipg           mail        pidof           rpm               stty        vim
chgrp                 deallocvt       fillup          kbd_mode      mapscrn     ping            sash              su          vim-normal
chmod                 df              find            kbdrate       md5sum      ping6           sed               sync        ypdomainname
chown                 dmesg           fsync           keyctl        mkdir       pkill           setfont           tar         zcat
chvt                  dnsdomainname   fuser           kill          mknod       ps              setkeycodes       tcsh        zsh
cp                    domainname      gawk            ksh           mktemp      psfaddtable     setleds           testutf8
cpio                  dumpkeys        getkeycodes     ksh93         more        psfgettable     setmetamode       touch
csh                   echo            grep            ln            mount       psfstriptable   setserial         true
date                  ed              guess_encoding  loadkeys      mountpoint  psfxtable       sh                umount
dbus-cleanup-sockets  egrep           gunzip          loadunimap    mv          pwd             showconsolefont   uname
chuck@linux-jgfu:/bin>
```

/sbin

- Similar to /bin
- Contains binary files not intended for the average computer user

/etc

- Contains configuration files, such as for web servers, boot loaders, security software, and many other applications

/etcinittab File

- Sets boot-up process and operation
  - Example: init level for the system on start-up

label	run_level	action:a	process
boot	bootwait	initdefault	sysinit

/dev

- Contains device files
  - Interfaces to devices
- All devices should have a device file in /dev
- Device naming conventions:
  - hd = hard drive
  - fd = floppy drive
  - cd = CD
  - Example: Main hard drive can be /dev/hd0

/mnt

- Many devices are mounted in /mnt
- Drives must be mounted prior to use
- Checking this directory lets you know what is currently mounted on system

/boot

- Contains files critical for booting
- Boot loader (LILO or GRUB) looks in this directory
- Kernel images commonly located in /boot

/usr

- Contains subdirectories for individual users

/var and /varspool

- /var
  - Contains data that is changed during system operation
- /varspool
  - Contains the print queue

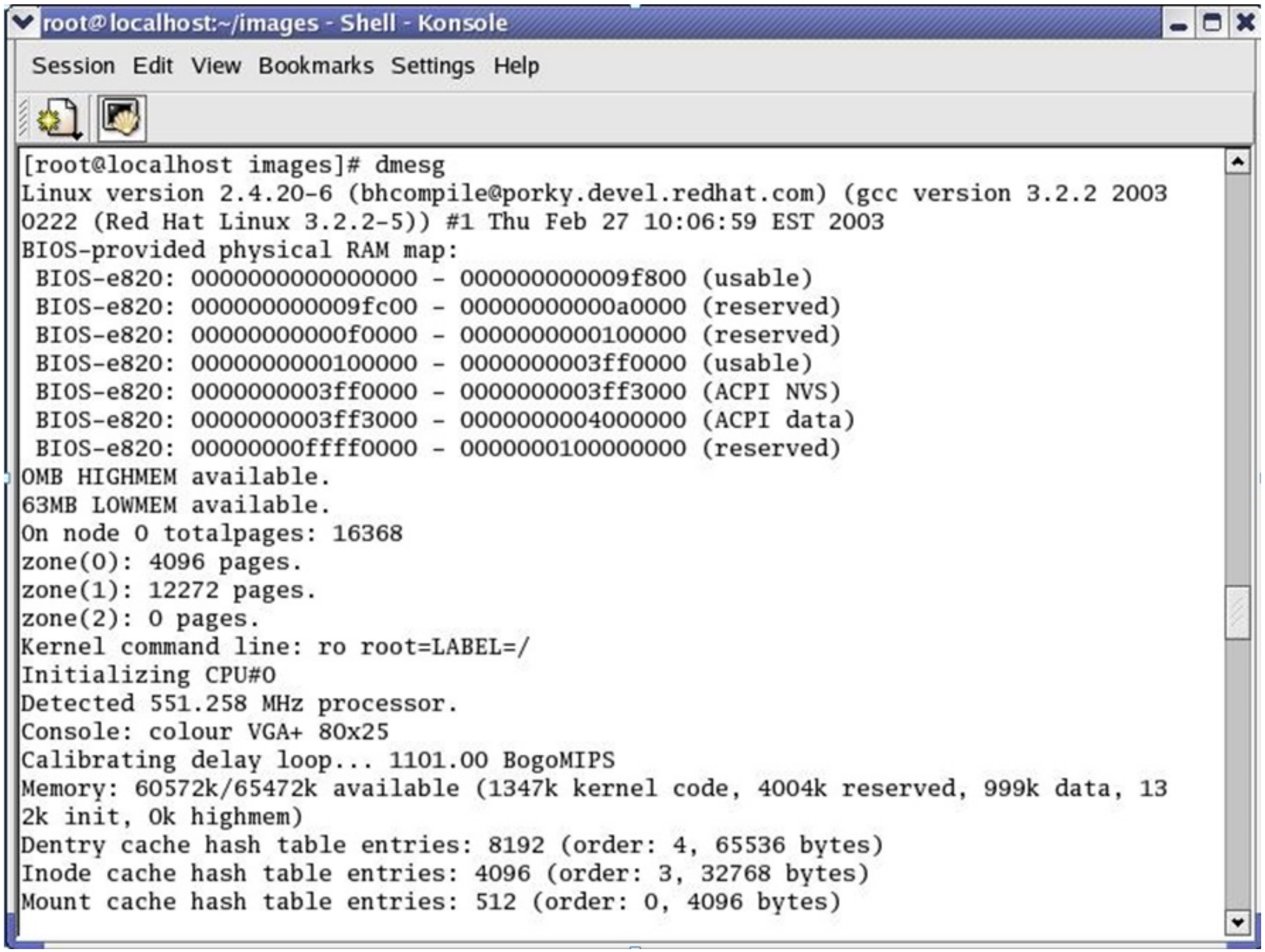
The /proc Directory

```
sys : bash
File  Edit  View  Scrollback  Bookmarks  Settings  Help
chuck@linux-jgfu:/proc> ls
1     1246  1720  20     2242  26    3     4061  556   asound     dri          kcore          misc           self           tty
10    13    1750  2021   2254  2613  339   4062  57    buddyinfo  driver       kdb            modules        slabinfo       uptime
1029  1343  1763  2052   229   2636  3584  4072  58    bus        execdomains  key-users      mounts         softirqs       version
1031  1370  18    21     23    28    3618  4222  59    cgroups    fb           kmsg           mtrr           splash         vmallocinfo
11    1377  181   21992  2355  2915  3638  4223  6     cmdline    filesystems  kpagecount     net            stat           vmstat
1116  1378  1901  21994  2357  2949  3687  4224  66    config.gz  fs           kpageflags     pagetypeinfo   swaps          zoneinfo
1154  14    1906  22     2395  2965  3741  4225  67    cpuinfo    interrupts   latency_stats  partitions     sys
1160  1478  1912  22044  24    2966  3750  4226  7     crypto     iomem        loadavg        reserve_info   sysrq-trigger
1163  15    1951  22145  2478  2969  3787  4227  8     devices    ioports      locks          sched_debug    sysvipc
12    16    1978  222    2489  2976  4     5     9     diskstats  irq          mdstat         schedstat      timer_list
120   17    2     223    25    2988  4036  542   acpi  dma        kallsyms     meminfo        scsi           timer_stats
chuck@linux-jgfu:/proc>
```

Shell Commands for Forensics

- Linux has hundreds of shell commands
- Some can be very useful in forensic investigations

The dmesg Command



The `pstree` Command

```
test@debian:~$ pstree
init─┬─NetworkManager───{NetworkManager}
     ├─NetworkManagerD
     ├─acpid
     ├─anacron───run-parts───apt───sleep
     ├─apache2───5*[apache2]
     ├─atd
     ├─avahi-daemon───avahi-daemon
     ├─bonobo-activati───{bonobo-activati}
     ├─cron
     ├─cupsd
     ├─2*[dbus-daemon]
     ├─dbus-launch
     ├─dhcdbd───dhclient
     ├─exim4
     ├─gconfd-2
     ├─gdm───gdm─┬─Xorg
     │           └─x-session-manag─┬─bluetooth-apple
     │                             ├─gnome-panel
     │                             ├─gnome-settings-───{gnome-settings-}
     │                             ├─kerneloops-appl
     │                             ├─metacity
     │                             ├─nautilus
     │                             ├─nm-applet
     │                             ├─seahorse-agent
     │                             ├─system-config-p
     │                             ├─update-notifier
     │                             └─{x-session-manag}
     ├─6*[getty]
     ├─gnome-keyring-d
     ├─gnome-power-man
     ├─gnome-screensav
     ├─gnome-terminal─┬─bash───pstree
     │                ├─gnome-pty-helpe
     │                └─{gnome-terminal}
     ├─gnome-vfs-daemo
     ├─gnome-volume-ma
     ├─hald───hald-runner─┬─hald-addon-acpi
     │                    ├─hald-addon-inpu
     │                    └─hald-addon-stor
     ├─kerneloops
     ├─mapping-daemon
     ├─mixer_applet2───{mixer_applet2}
     ├─notification-da
     ├─portmap
     ├─postgres───4*[postgres]
     ├─rpc.statd
     ├─rsyslogd───2*[{rsyslogd}]
```

The `file` Command

```
testuser : bash <2>
File  Edit  View  Scrollback  Bookmarks  Settings  Help
testuser@linux-58ka:~> ls
args.2sh         Documents      myscript.2sh~   New Folder   test
args.sh~         Downloads      myscript.2sh~~  Pictures     test.txt
bin              math.sh~       myscript2.sh~   Public       triangle.sh~
ChuckEasttom.jpg Music          myscript3.sh~   public_html  Videos
Desktop          myscript.2sh   myscript.sh~    Templates
testuser@linux-58ka:~> file ChuckEasttom.jpg
ChuckEasttom.jpg: JPEG image data, JFIF standard 1.02
testuser@linux-58ka:~> file test.txt
test.txt: ASCII text, with no line terminators
testuser@linux-58ka:~> 
```

Undeleting Linux Files: Manually

Move system to single-user mode

Use `grep` or similar command

Example: `grep -b 'search-text' /dev/partition > file.txt`

Use command-line editor to view file

Let's Play:

Identify the Shell Command

Command 1

- Displays the commands that have previously been entered

Answer choices:

a. dmesg
b. grep
c. history
d. ls

Answer 1

history

Command 2

- Shows all the processes in the form of a tree structure

Answer choices:

a. ps
b. pstree
c. ls
d. top

Answer 2

pstree

Command 3

- Takes the name you provide and returns the ID for that process; can work with partial names

Answer choices:

a. pgrep
b. dd
c. grep
d. file

Answer 3

pgrep

Command 4

- Lists the processes in the order of how much CPU time the process is utilizing

Answer choices:

a. ps
b. ls
c. su
d. top

Answer 4

top

Command 5

- A criminal changes a file extension. This command can identify the file.

Answer choices:

a. history
b. ls
c. file
d. mount

Answer 5

file

Command 6

- Halts a running process based on the process ID (PID) you provide

Answer choices:

a. kill
b. dmesg
c. ps
d. finger

Answer 6

kill

Command 7

- Invokes the super user mode

Answer choices:

a. who
b. grep
c. finger
d. su

Answer 7

su

Command 8

- Provides information about a specific user

Answer choices:

a. finger
b. who
c. su
d. grep

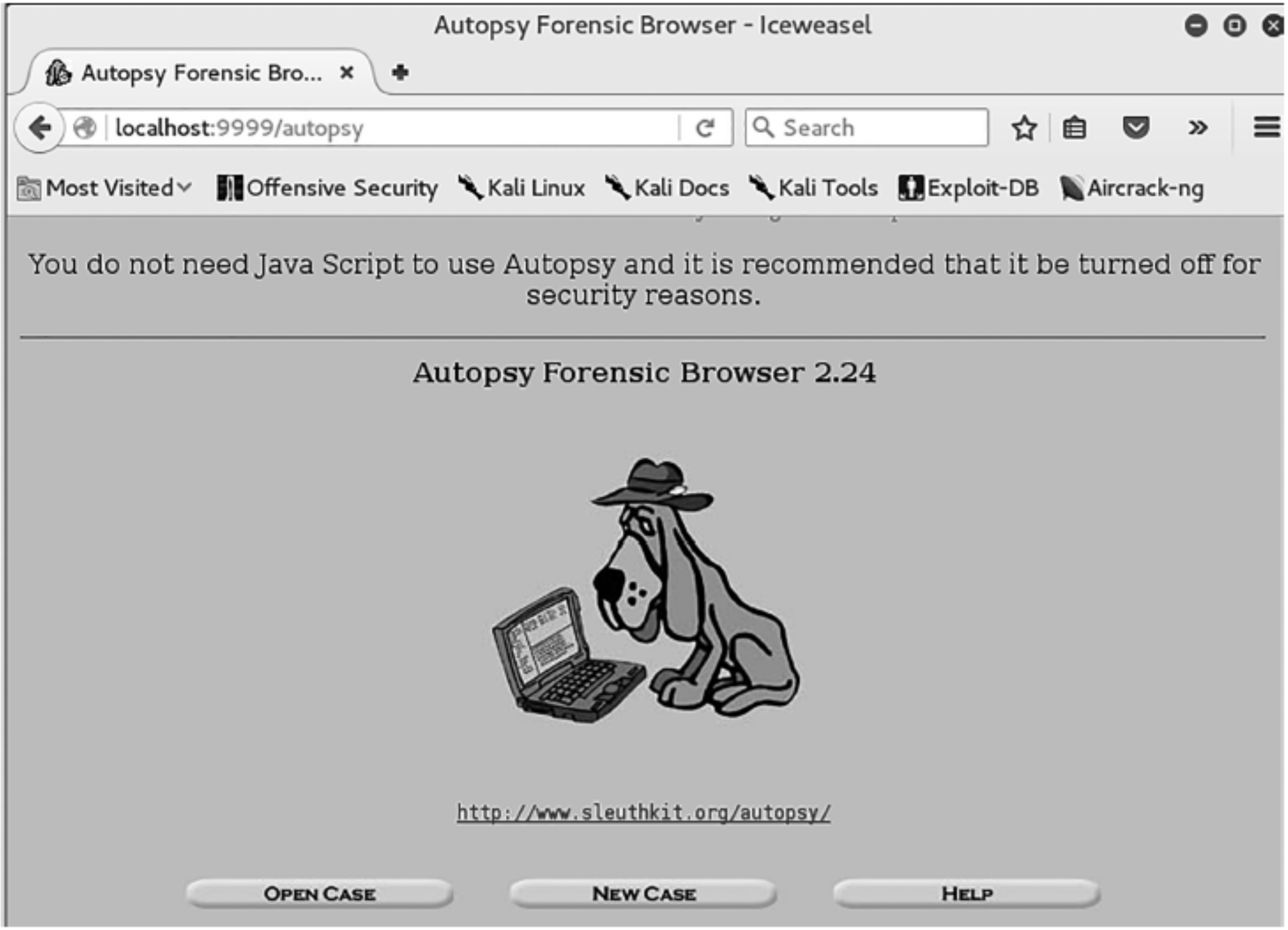
Answer 8

finger

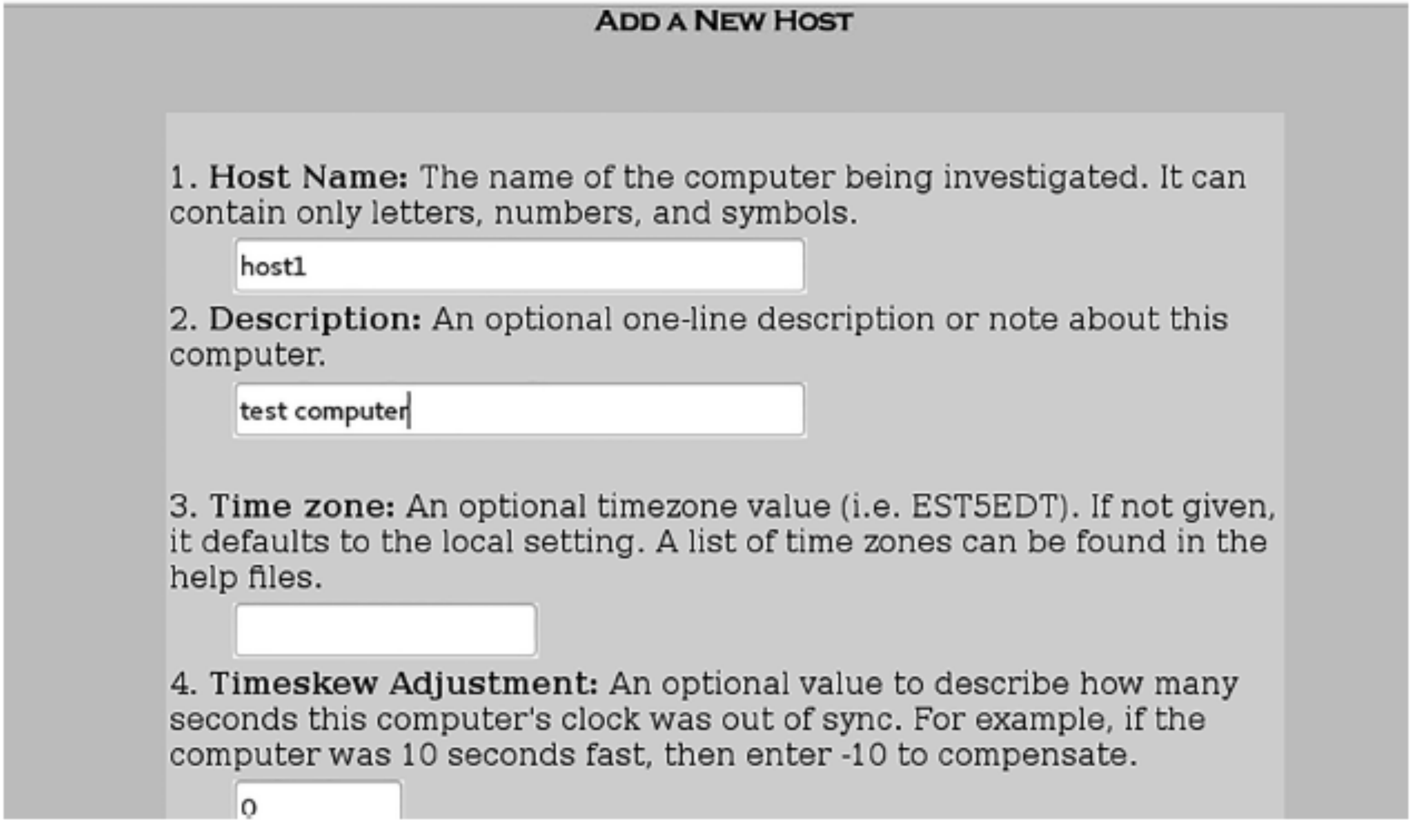
Kali Linux

- Has a number of forensics tools
- Can use as quality control tool to complement OSForensics, FTK, or Encase
- Includes Autopsy, a web-based graphical user interface for the command-line tool Sleuth Kit

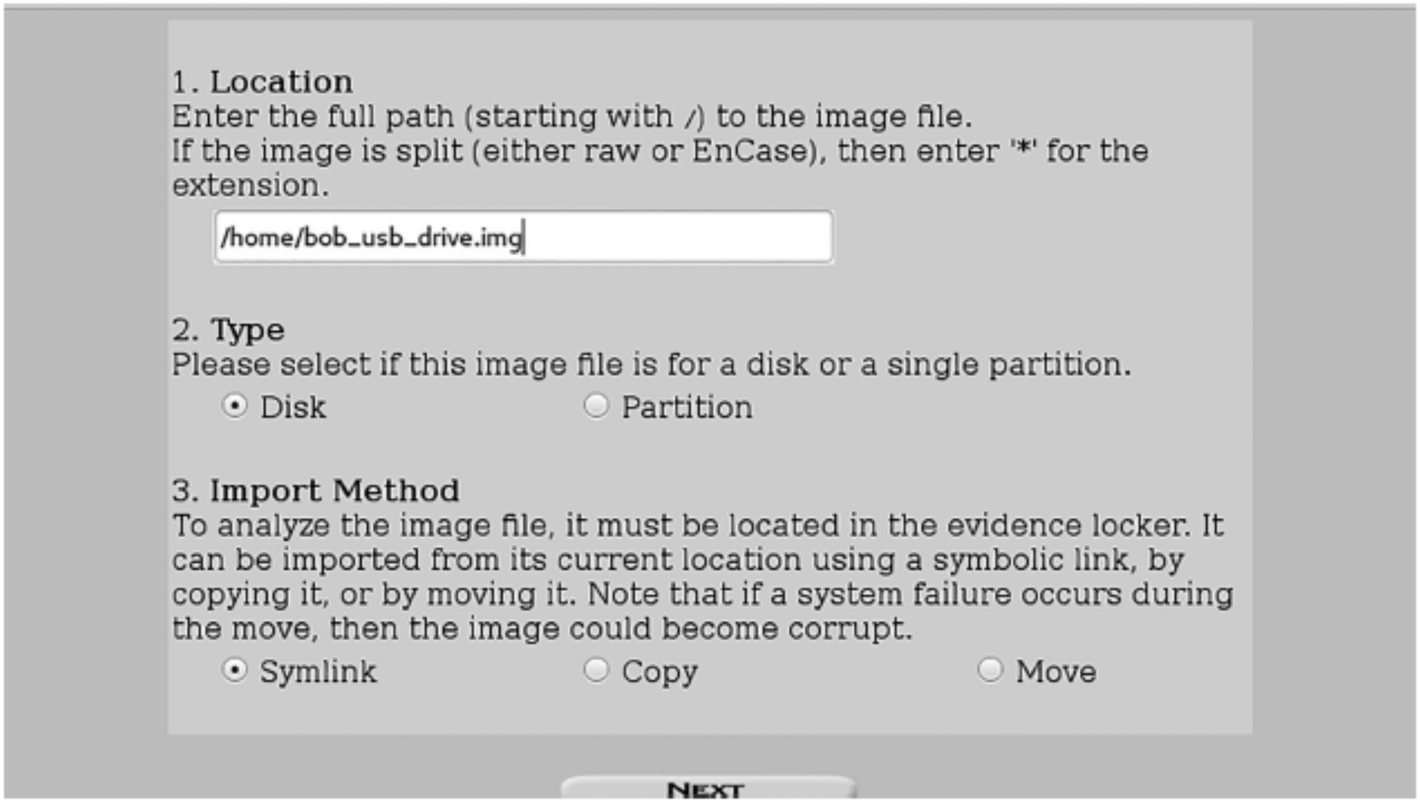
Autopsy


Autopsy Forensic Browser - Iceweasel
Autopsy Forensic Bro...
localhost:9999/autopsy
Search
Most Visited
Offensive Security
Kali Linux
Kali Docs
Kali Tools
Exploit-DB
Aircrack-ng
You do not need Java Script to use Autopsy and it is recommended that it be turned off for security reasons.
Autopsy Forensic Browser 2.24
http://www.sleuthkit.org/autopsy/
OPEN CASE
NEW CASE
HELP

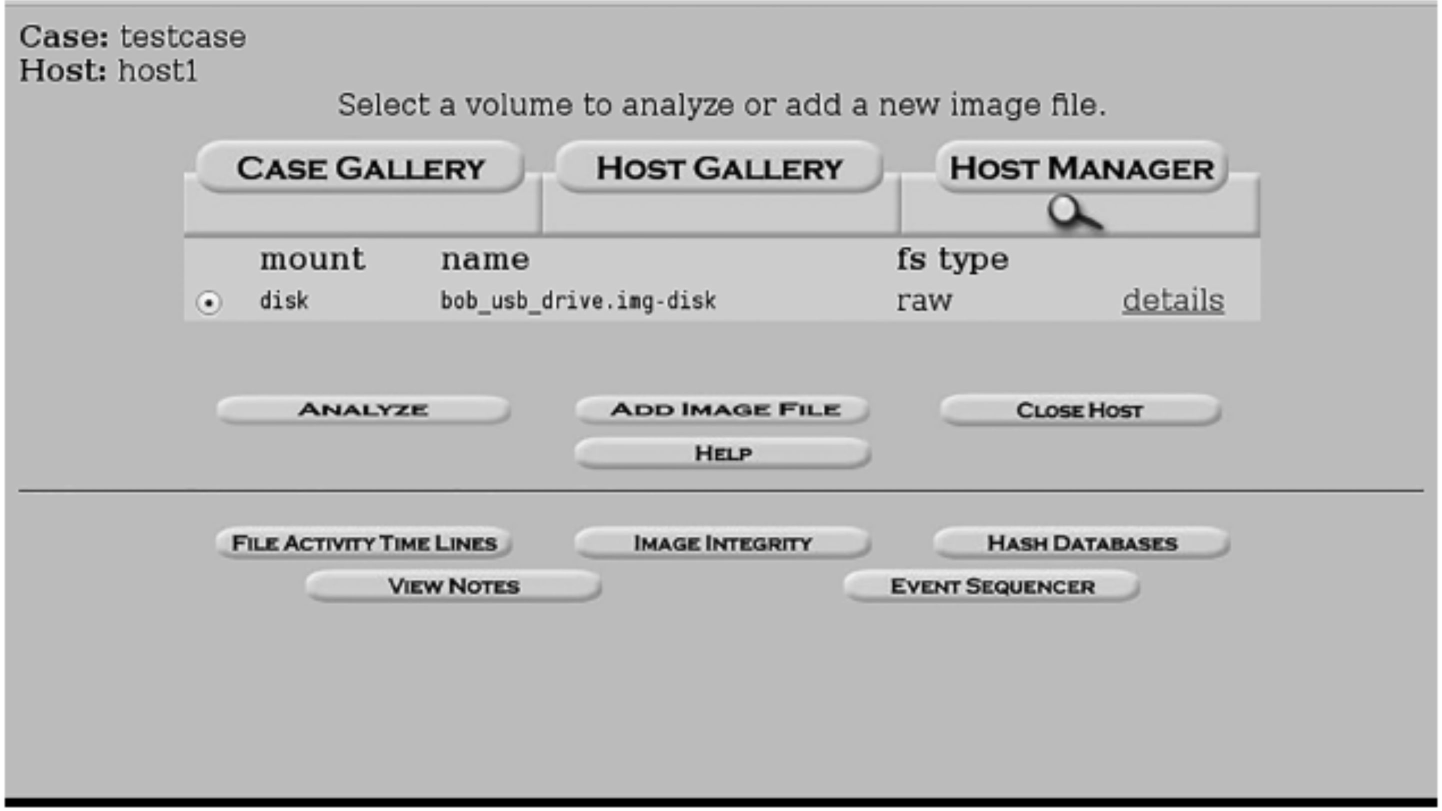
Autopsy (Cont.)



Autopsy (Cont.)



Autopsy (Cont.)



Summary

- Linux file systems
- What to look for in Linux system logs
- Forensically interesting Linux directories
- Important Linux shell commands
- How to undelete files from Linux

THANK YOU

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